Two-period pricing with symmetric switching preferences

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1 Model, histories, and normalization

One firm sells one inside product j against outside option 0 in periods $t = 1, 2$. Type r is fixed and observed by the firm, which prices separately for that type. All formulas below are conditional on r; its subscript and the fixed product index are suppressed. There is no pooling of heterogeneous types. The initial realized choice is $x_0 \in \{j, 0\}$. The second-period price can depend on the realized first-period choice x_1. Independent Type-I extreme-value shocks enter both alternatives in each period.

Before normalization the deterministic utilities are

$$V_{jt}^{\text{raw}} = \beta + \alpha p_t + \gamma \mathbf{1}\{x_{t-1} = 0\}, \qquad V_{0t}^{\text{raw}} = \gamma \mathbf{1}\{x_{t-1} = j\}.$$

The payoff to switching applies in either direction. Subtracting the raw outside utility from both alternatives gives normalized outside utility zero and

$$v_t(h) = \beta + \alpha p_t(h) + q(h)\gamma, \qquad q(j) = -1, \quad q(0) = +1.$$

Normalization retains the outside switching payoff through the signed inside utility difference; it does not delete that payoff. Negative γ denotes switching aversion and positive γ denotes novelty preference.

$$s_t(h) = \frac{e^{v_t(h)}}{1 + e^{v_t(h)}}, \quad s_{0t}(h) = 1 - s_t(h), \quad \partial_p s_t = \alpha s_t(1 - s_t), \quad (\partial_\gamma s_t)_p = q(h) s_t(1 - s_t).$$

A realized state x_t, a hypothetical switching indicator, and the probability s_t are distinct objects. Every derivative below holds the discrete history fixed.

Assumptions. $\alpha < 0$, $0 < \delta < 1$, fixed marginal cost c, finite parameters, and interior binary-logit shares. Prices are unrestricted real numbers: introductory subsidies are allowed. Continuation values are conditional on realized states and have no direct dependence on p_1. Type does not evolve. The case $\delta = 0$ is used only as a static counterfactual.

Notation	Meaning
$q = q(x_0)$	Signed initial-state coefficient
$s = s_1^*(x_0), \quad r_0 = 1 - s$	First-period equilibrium shares
$u = s_2^*(j), \quad v = s_2^*(0)$	Inside shares in the two continuation states
$V_j = V_2(j), \quad V_0 = V_2(0)$	Optimized continuation profits
$\Delta V = V_j - V_0, \quad B = u + v$	Continuation difference and sum of terminal shares
Π^*	Expected discounted profit per initial consumer

2 Backward induction and optimality

For each realized first-period choice $a \in \{j, 0\}$, solve

$$V_2(a) = \max_{p_2(a)} s_2(a)[p_2(a) - c].$$

Its first-order condition is

$$0 = s_2 + \alpha s_2(1 - s_2)(p_2 - c), \qquad p_2^*(a) = c - \frac{1}{\alpha[1 - s_2^*(a)]}.$$

This is an implicit equation: the share depends on the same price being solved. Both V_j and V_0 must be computed before period 1.

First-period profit integrates over both possible realized histories:

$$\begin{aligned}\Pi(p_1; x_0) &= s_1(p_1 - c) + \delta[s_1 V_j + (1 - s_1)V_0] \\ &= \delta V_0 + s_1[p_1 - c + \delta \Delta V].\end{aligned}$$

Because the branch values do not depend directly on p_1,

$$\partial_{p_1}\{\delta[s_1 V_j + (1 - s_1)V_0]\} = \delta\alpha s_1(1 - s_1)(V_j - V_0).$$

Consequently,

$$0 = s_1 + \alpha s_1(1 - s_1)[p_1 - c + \delta\Delta V], \qquad p_1^* = c - \frac{1}{\alpha(1 - s)} - \delta\Delta V.$$

The static FOC component $c - 1/[\alpha(1 - s)]$ uses the share at the dynamic optimum. It is not the counterfactual static optimum, which must be solved separately with $\delta = 0$.

Proposition 1 (Unique optimal prices)**.** *Under the stated assumptions each continuation price and the first-period price has a unique finite global optimum, with no lower bound imposed on* p_1*.*

Proof. Both price problems reduce, up to a constant, to maximizing $s(p)(p - k)$ with $s(p) = \text{logit}^{-1}(b + \alpha p)$. Use $k = c$ in period 2 and $k = c - \delta\Delta V$ in period 1. The sign of the profit derivative is the sign of

$$H(p) = 1 + e^{b+\alpha p} + \alpha(p - k), \quad H'(p) = \alpha(1 + e^{b+\alpha p}) < 0.$$

H ranges from positive infinity to negative infinity, so its root is unique and the objective increases before it and decreases afterward. The price may lie below physical cost c in period 1 when continuation incentives justify it. □

3 Envelope derivatives and conditional comparative statics

Differentiating each continuation value using the envelope theorem gives

$$V_2'(a) = q(a)s_2(1 - s_2)(p_2^* - c) = -\frac{q(a)s_2}{\alpha}.$$

Implicit differentiation of the terminal FOC gives the same expression for the optimal price derivative:

$$p_2^{*\prime}(a) = -\frac{q(a)s_2}{\alpha}, \quad V_j' = u/\alpha, \quad V_0' = -v/\alpha, \quad \Delta V' = (u+v)/\alpha.$$

In particular, $p_2^{*\prime}(j) < 0$ and $p_2^{*\prime}(0) > 0$. The total equilibrium terminal-share derivative is $s_2' = q(a)s_2(1-s_2)^2$, including the optimal-price response.

For the first-period implicit pricing equation let $F = p_1 - c + 1/[\alpha(1-s_1)] + \delta\Delta V$. At fixed history,

$$F_{p_1} = \frac{1}{r_0}, \qquad F_\gamma = \frac{qs}{\alpha r_0} + \frac{\delta B}{\alpha}, \qquad \boxed{p_1^{*\prime} = -\frac{qs + \delta r_0 B}{\alpha}}.$$

The share therefore changes according to

$$s' = sr_0(q + \alpha p_1^{*\prime}) = sr_0^2(q - \delta B).$$

The first-period envelope theorem retains both continuation branches:

$$\begin{aligned}\Pi^{*\prime} &= qsr_0[p_1^* - c + \delta\Delta V] + \delta[sV_j' + r_0V_0'] \\ &= \boxed{\frac{-qs + \delta(su - r_0v)}{\alpha}}.\end{aligned}$$

The term $\delta V_0'$ must not be dropped. The envelope partial derivative holds the optimizing price fixed inside the partial derivative; it does not hold the continuation value fixed as γ varies.

For the analytical second price derivative define

$$u' = -u(1-u)^2, \quad v' = v(1-v)^2, \quad B' = u' + v'.$$

Differentiating the first derivative and substituting s' gives

$$\boxed{p_1^{*\prime\prime} = -\frac{sr_0^2(q - \delta B)^2 + \delta r_0 B'}{\alpha}}.$$

This expression is used analytically in checks, rather than obtained from noisy twice-differenced optimized prices. No second-derivative figure is included.

Proposition 2 (Conditional signs)**.** *For an initially outside consumer, $p_1^{*\prime} > 0$ and $\Pi^{*\prime} > 0$. For an initially inside consumer,*

$$p_1^{*\prime} > 0 \iff \delta r_0(u+v) > s, \qquad \Pi^{*\prime} > 0 \iff \delta r_0 v > s(1+\delta u).$$

Equalities give zero derivatives; reversed inequalities give negative derivatives.

Proof. For $q = 1$, the price numerator $s + \delta r_0 B$ is positive and $-1/\alpha > 0$. The profit numerator is $-s(1-\delta u) - \delta r_0 v < 0$ because $\delta u < 1$; dividing by $\alpha < 0$ makes the derivative positive. Substituting $q = -1$ gives the two stated thresholds. These statements do not establish a common monotonicity sign for both histories or a global sign for the price ratios. □

4 Price ratios: three distinct conditioning statements

The state-matched ratios are descriptive comparisons:

$$R_j = \frac{p_1^*(x_0 = j)}{p_2^*(x_1 = j)}, \qquad R_0 = \frac{p_1^*(x_0 = 0)}{p_2^*(x_1 = 0)}.$$

They do not necessarily describe one realized consumer path. The acquisition comparison is instead

$$R_{0\to j} = \frac{p_1^*(x_0 = 0)}{p_2^*(x_1 = j)},$$

conditional on acquisition in period 1. It is not an expectation over histories. For each definition use its own numerator, denominator and state derivatives:

$$\boxed{R' = \frac{p_2^* p_1^{*\prime} - p_1^* p_2^{*\prime}}{(p_2^*)^2}}.$$

Terminal prices exceed c; the benchmark $c = 0$ therefore has strictly positive ratio denominators. For other calibrations a ratio is only defined when its denominator is nonzero. No expected ratio or ratio of expectations is substituted.

5 Numerical benchmark and verification

Every plotted point reoptimizes both terminal branches and period 1:

$$\beta = 2, \qquad \alpha = -1, \qquad \delta = 0.92, \qquad c = 0, \qquad \gamma \in [-3, 3] \text{ on a 601-point grid.}$$

The calibration is illustrative, not estimated. SymPy verifies raw-utility normalization, both branch weights, the profit-derived FOCs, implicit price derivatives and the envelope identities. Numerical checks compare all 1,202 benchmark state/grid solutions with the handoff's independent reference. Central differences reoptimize the full model at nearby parameters; an independent scalar profit optimizer checks optimality. Additional calibrations check derivatives and an introductory-subsidy case. Numerical evidence is not a proof of unqualified global comparative-static signs.

Selected optimized values

x_0	γ	p_1^*	$p_2^*(j)$	$p_2^*(0)$	Π^*	$\Pi^{*\prime}$
j	-3	3.395	3.926	1.120	5.087	-1.387
j	-1	2.236	2.557	1.567	2.669	-0.959
j	0	2.000	2.000	2.000	1.920	-0.500
j	1	2.209	1.567	2.557	1.731	0.125
j	3	3.592	1.120	3.926	2.702	0.668
0	-3	-0.780	3.926	1.120	0.913	0.195
0	-1	1.045	2.557	1.567	1.478	0.385
0	0	2.000	2.000	2.000	1.920	0.500
0	1	2.956	1.567	2.557	2.478	0.615
0	3	4.801	1.120	3.926	3.912	0.804

6 Symbolically derived equations

Here $q_2 = q(x_1)$, $s_2 = s_2(x_1)$ and $p_2 = p_2(x_1)$ describe a fixed terminal history. Price equations and comparative-static right-hand sides use equilibrium shares; the profit and FOCs use shares at the candidate prices.

$$v_1 = \alpha p_1 + \beta + \gamma q$$

$$v_2 = \alpha p_2 + \beta + \gamma q_2$$

$$\Pi = \delta \left(V_{20} \left(1 - s\right) + V_{2j} s\right) + s \left(-c + p_1\right)$$

$$F_1 = \alpha s \left(1 - s\right) \left(-c + \delta \left(-V_{20} + V_{2j}\right) + p_1\right) + s$$

$$F_2 = \alpha s_2 \left(1 - s_2\right) \left(-c + p_2\right) + s_2$$

$$p_1^* = c - \delta \left(-V_{20} + V_{2j}\right) - \frac{1}{\alpha \left(1 - s\right)}$$

$$p_2^* = c - \frac{1}{\alpha \left(1 - s_2\right)}$$

$$\frac{dp_1^*}{d\gamma} = \frac{\delta \left(s - 1\right) \left(u + v\right) - qs}{\alpha}$$

$$\frac{dp_2^*}{d\gamma} = -\frac{q_2 s_2}{\alpha}$$

$$\frac{d\Pi^*}{d\gamma} = \frac{\delta \left(su + v \left(s - 1\right)\right) - qs}{\alpha}$$

State-matched price-ratio derivatives

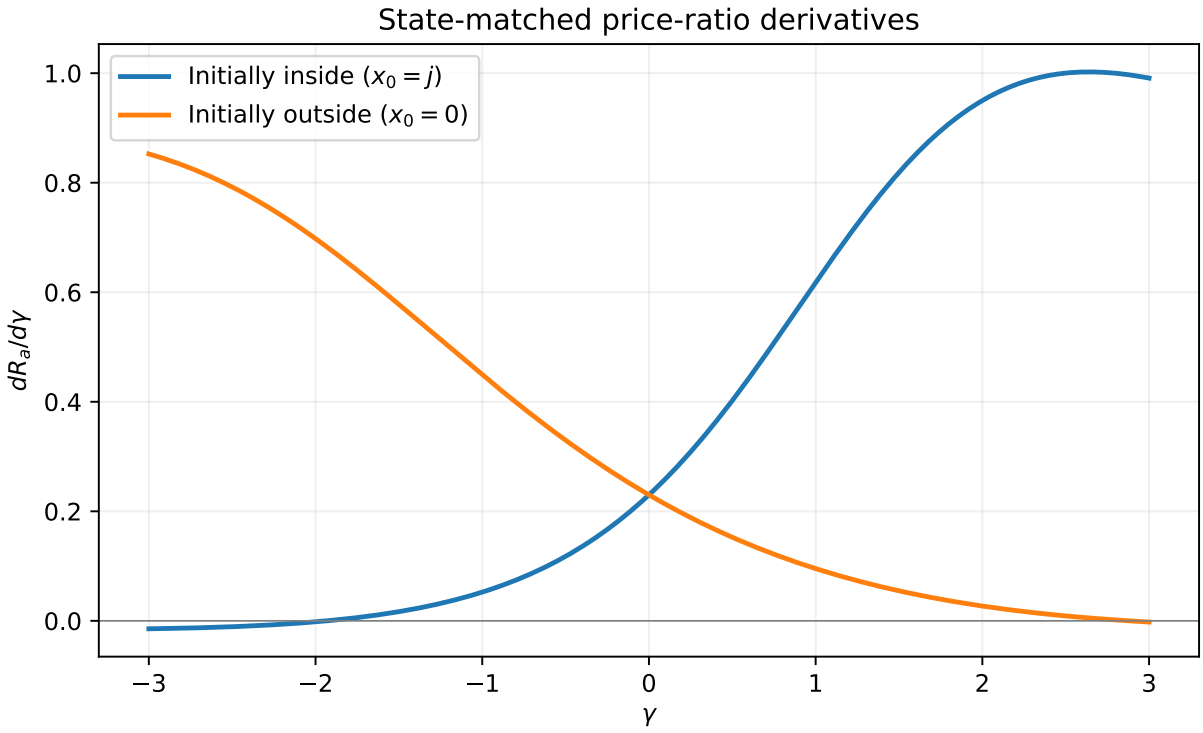


Figure 1: $R_j = p_1^*(j)/p_2^*(j)$ and $R_0 = p_1^*(0)/p_2^*(0)$. Each curve holds its stated histories fixed and uses the analytical quotient rule. Benchmark: $\beta = 2$, $\alpha = -1$, $\delta = 0.92$, $c = 0$; $\gamma \in [-3, 3]$.

Induce-then-harvest price-ratio derivative

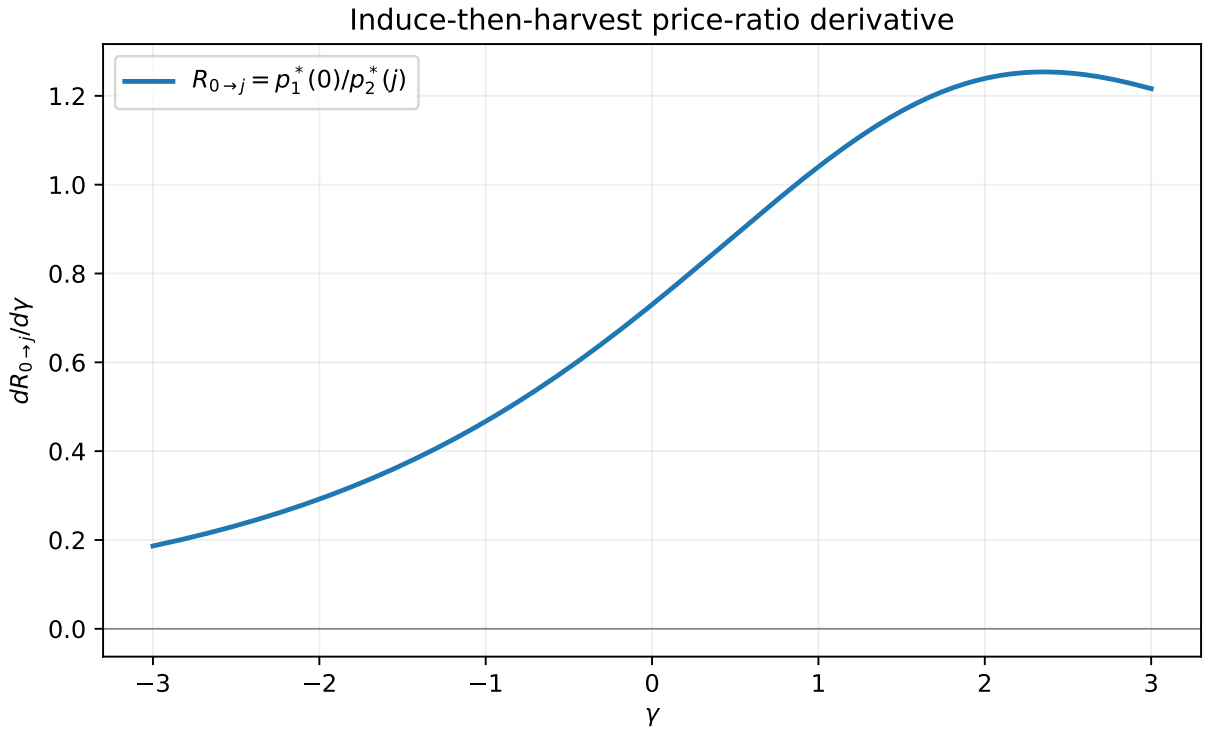


Figure 2: $R_{0\to j} = p_1^*(0)/p_2^*(j)$, conditional on a first-period acquisition. This is a path-specific comparison, not an ex-ante expectation. Benchmark: $\beta = 2$, $\alpha = -1$, $\delta = 0.92$, $c = 0$; $\gamma \in [-3, 3]$.

Optimized profit levels

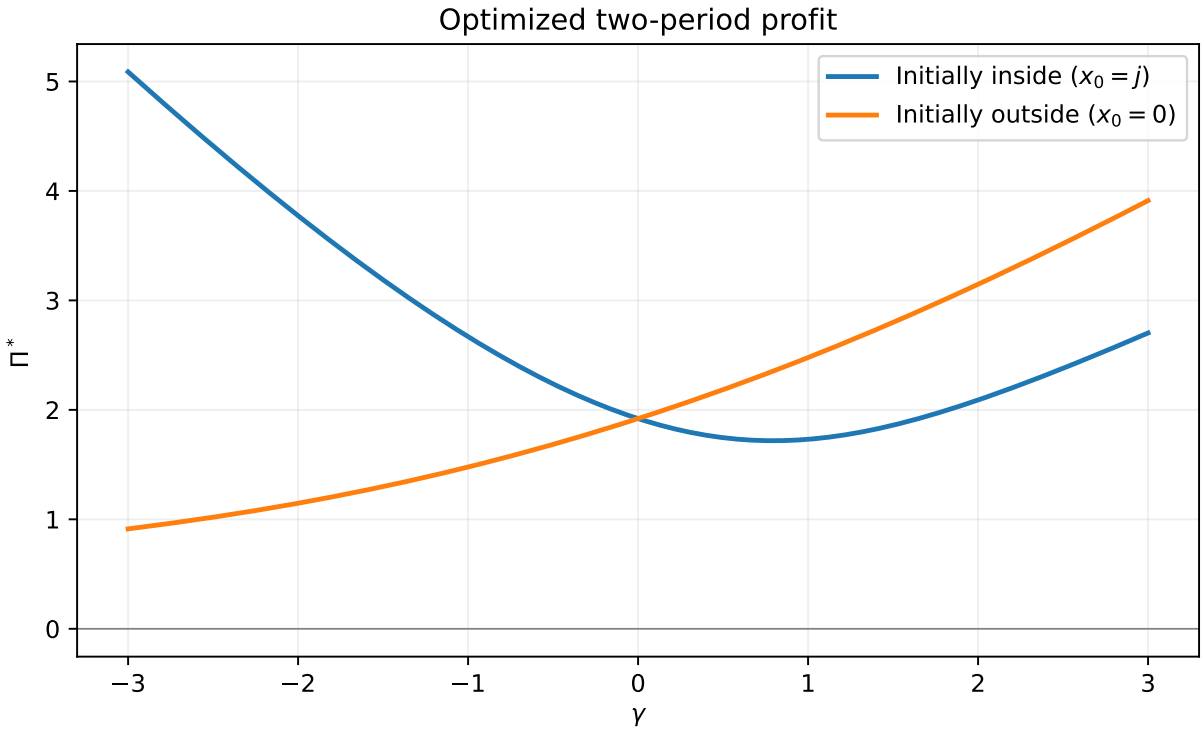


Figure 3: Expected discounted profit includes both realized continuation histories. The two curves condition on the initial realized state. Benchmark: $\beta = 2$, $\alpha = -1$, $\delta = 0.92$, $c = 0$; $\gamma \in [-3, 3]$.

Optimized profit derivative

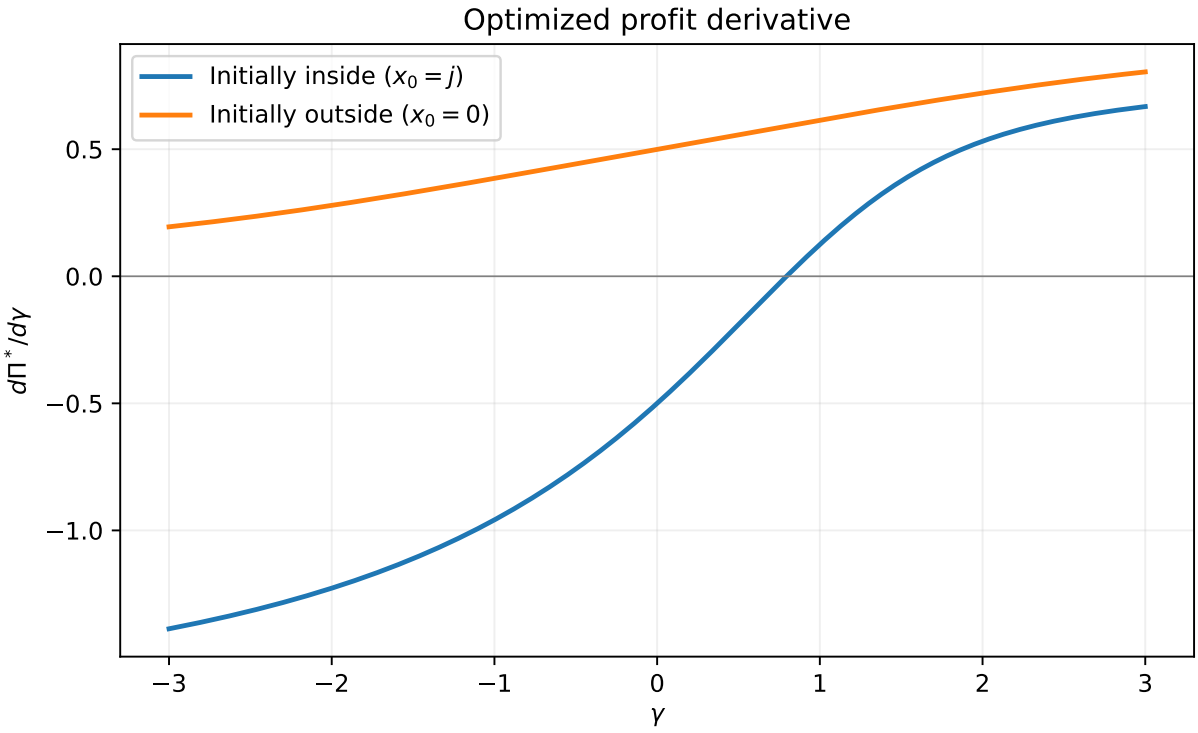


Figure 4: Total derivatives use the envelope theorem and both continuation-value derivatives. Their signs depend on the initial history and the stated thresholds. Benchmark: $\beta = 2$, $\alpha = -1$, $\delta = 0.92$, $c = 0$; $\gamma \in [-3, 3]$.

Optimal prices in both periods

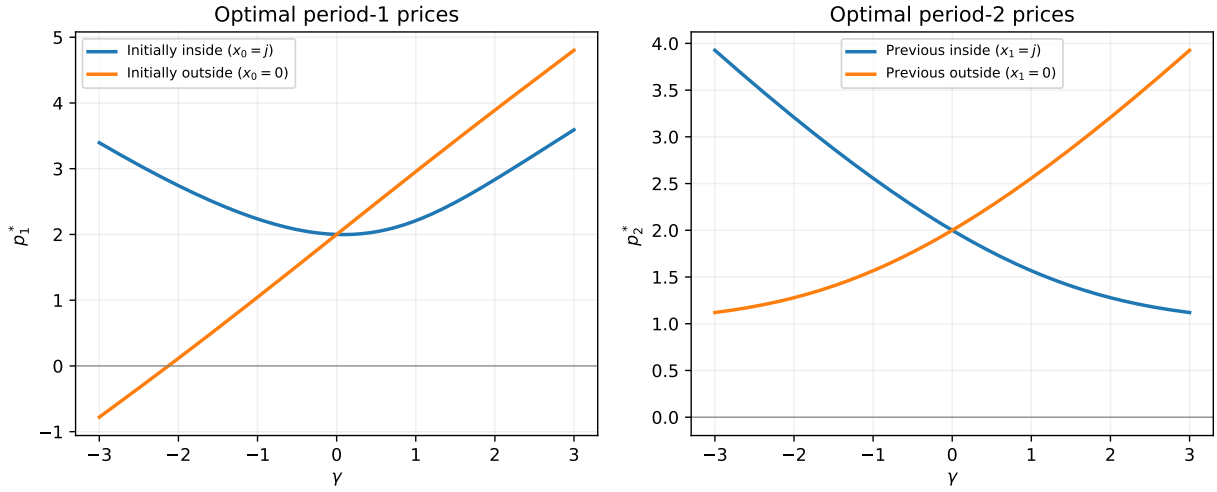


Figure 5: Left: period-1 prices for both initial states. Right: period-2 prices for both realized first-period states. Prices are unconstrained; all shares adjust. Benchmark: $\beta = 2$, $\alpha = -1$, $\delta = 0.92$, $c = 0$; $\gamma \in [-3, 3]$.

Optimal period-1 price derivative

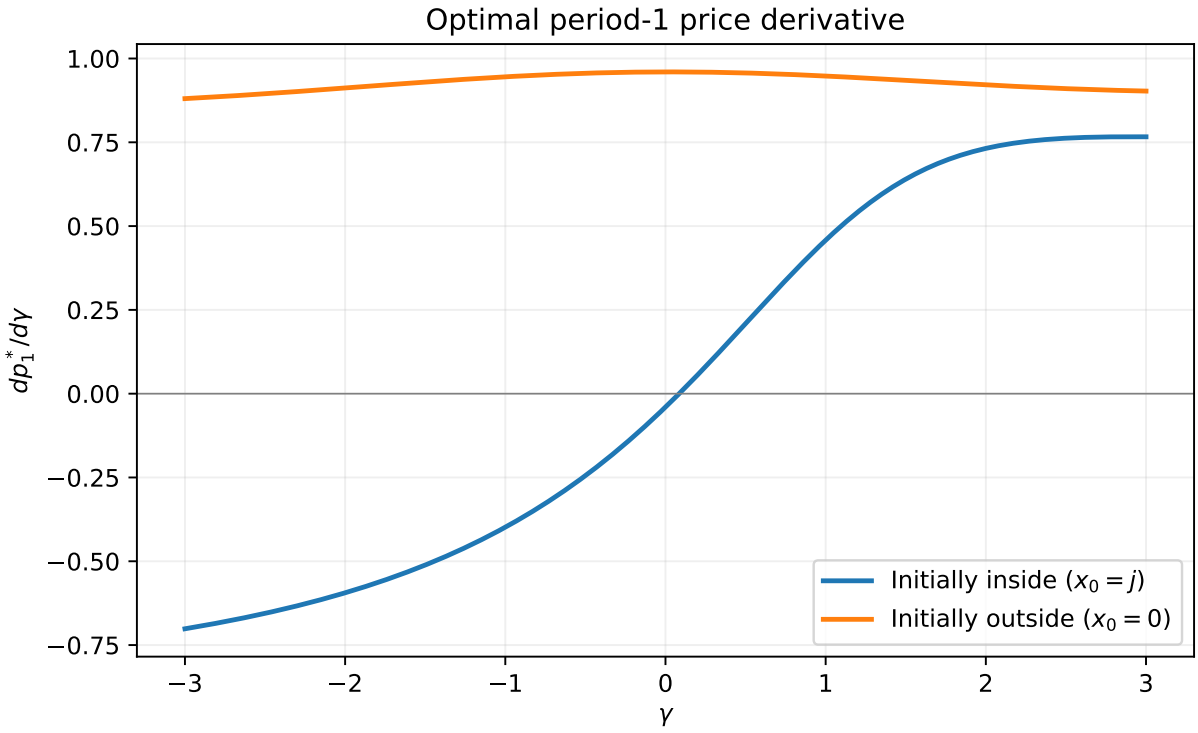


Figure 6: Analytical total derivatives, with both continuation branches reoptimized. Initial-state signs are interpreted using the proved conditional statements. Benchmark: $\beta = 2$, $\alpha = -1$, $\delta = 0.92$, $c = 0$; $\gamma \in [-3, 3]$.